Cybersecurity Internship Final Report - Complete Security Hardening Project

Intern Information

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Project Title: User Management System – Full Security Hardening Lifecycle
Duration: 6 Weeks (June – July 2026)
Organization: DevelopersHub
Department: Security Operations / Cybersecurity

Executive Summary

This report presents the complete 6-week cybersecurity internship journey focused on securing a **User Management System** through a full security hardening lifecycle. The project encompassed every stage of the cybersecurity process—from initial vulnerability assessment to final deployment and audit. The primary objective was to transform a vulnerable web application into a hardened, production-ready system aligned with **OWASP Top 10** and industry best practices.

The internship involved identifying and mitigating vulnerabilities such as **SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF)**, implementing **security headers**, enforcing **rate limiting**, and conducting **ethical hacking** and **security audits** using professional tools including **OWASP ZAP, Burp Suite, Nikto, SQLMap, and Lynis**.

By the end of the internship, the system achieved a significant improvement in overall security posture, with measurable enhancements in compliance, authentication, and resilience against common web attacks.

Project Overview

The **User Management System Security Hardening Project** aimed to secure a web-based application that manages user registration, authentication, and role-based access. The project followed a structured six-week plan, each week focusing on a specific phase of the cybersecurity lifecycle.

Project Objectives

- Conduct a full vulnerability assessment and penetration test.
- Implement secure coding practices and mitigate identified vulnerabilities.
- Strengthen authentication, authorization, and session management mechanisms.
- Apply web security headers and rate limiting to prevent abuse.
- Perform ethical hacking to validate security controls.
- Conduct final audits and prepare the system for production deployment.

Weekly Progress and Achievements

Week 1: Security Assessment & Vulnerability Identification

- Conducted reconnaissance using **Nmap**, **WhatWeb**, and **Nikto** to identify open ports, technologies, and potential misconfigurations.
- Performed vulnerability scanning using **OWASP ZAP** and **Burp Suite**.
- Discovered critical vulnerabilities including **SQL Injection**, **XSS**, and **CSRF**.
- Documented all findings in a vulnerability assessment report.
- Established baseline security metrics for comparison.

Key Deliverables:

- Initial vulnerability report
- Risk classification matrix
- Baseline security scorecard

Week 2: Critical Vulnerability Fixes (SQLi, XSS, CSRF)

- Implemented **prepared statements** to prevent SQL Injection.
- Added **Content Security Policy (CSP)** and input sanitization to mitigate XSS.
- Integrated **CSRF token validation** using secure, random tokens stored in **HttpOnly cookies**.
- Conducted retesting using **SQLMap** and **Burp Suite** to confirm remediation.
- Updated documentation to reflect fixed vulnerabilities.

Key Deliverables:

- Secure database interaction module
- Updated web forms with CSRF protection
- Post-fix vulnerability validation report

Week 3: Advanced Security Improvements (Logging & Authentication)

- Integrated **JWT-based authentication** with **bcrypt password hashing** for secure credential management.
- Implemented **Winston-based logging system** for centralized event tracking and incident response.
- Enhanced session management with **SameSite=Strict** and **Secure** cookie attributes.
- Configured **rate limiting** (5 login attempts per 15 minutes) to prevent brute-force attacks.
- Conducted internal code review for authentication and logging modules.

Key Deliverables:

- Secure authentication and session management system
- Logging and monitoring dashboard
- Rate limiting configuration

Week 4: Threat Detection & Web Security Hardening

- Applied advanced **security headers** including:
  - **Content-Security-Policy (CSP)**
  - **Strict-Transport-Security (HSTS)**
  - **X-Frame-Options**
  - **X-Content-Type-Options**
  - **Referrer-Policy**
  - **Permissions-Policy**
- Disabled **TRACE** method in Apache configuration to prevent cross-site tracing attacks.
- Hid PHP version and server information to prevent information disclosure.
- Conducted **Lynis** system hardening audit and remediated identified weaknesses.

Key Deliverables:

- Hardened Apache and PHP configuration
- Security headers implementation report
- Updated system hardening checklist

Week 5: Ethical Hacking & Exploiting Vulnerabilities

- Performed controlled penetration testing using **Kali Linux**, **Burp Suite**, and **OWASP ZAP**.
- Validated the effectiveness of implemented security controls.
- Simulated brute-force, XSS, and CSRF attacks to test resilience.
- Documented all test cases, results, and mitigation verification.
- Conducted peer review and internal audit of ethical hacking results.

Key Deliverables:

- Ethical hacking report
- Exploitation and mitigation validation logs
- Updated risk assessment

Week 6: Advanced Security Audits & Final Deployment

- Conducted final **security audit** using **Lynis** and **OWASP ZAP**.
- Verified compliance with **OWASP Top 10** and internal security policies.
- Prepared final documentation including audit reports, configuration files, and code repository.
- Deployed the hardened system to the production environment.
- Presented final results and metrics to the cybersecurity team.

Key Deliverables:

- Final security audit report
- Deployment documentation
- Project presentation and summary

Technical Implementation Summary

Security Feature	Implementation Details
Security Headers	CSP, HSTS, X-Frame-Options, X-Content-Type-Options, Referrer-Policy, Permissions-Policy
Rate Limiting	5 login attempts per 15 minutes
Authentication	JWT-based authentication with bcrypt password hashing
Input Validation	Prepared statements for SQL injection prevention
CSRF Protection	Token-based validation with HttpOnly cookies
Session Management	Secure session handling with SameSite=Strict
Logging	Winston-based centralized logging system

Tools and Technologies Used

Category	Tools
Vulnerability Assessment	Nmap, WhatWeb, Nikto
Penetration Testing	SQLMap, Burp Suite, OWASP ZAP
System Hardening	Lynis
Development Environment	XAMPP, DVWA, Kali Linux, Windows 11
Version Control	GitHub
Repository	https://github.com/mtalha27709-coder/user-management-security-hardening

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Security Metrics (Before vs After)

Metric	Before	After
Security Headers Grade	F	A+
Login Security	0/10	9/10
OWASP Compliance	20%	95%
Brute-Force Protection	❌	✅
SQL Injection Protection	❌	✅
CSRF Protection	❌	✅

Vulnerabilities Fixed

Vulnerability	Severity	Resolution
SQL Injection	Critical	Implemented prepared statements
Cross-Site Request Forgery (CSRF)	High	Added token-based validation
Cross-Site Scripting (XSS)	Critical	Applied CSP and input validation
TRACE Method Enabled	Medium	Disabled in Apache configuration
Information Disclosure	Low	Hidden PHP version and server info
Clickjacking	Medium	Added X-Frame-Options header

Results and Analysis

The project achieved a complete transformation of the User Management System's security posture. All critical vulnerabilities were mitigated, and the system now adheres to modern web security standards. The implementation of layered defenses, secure coding practices, and continuous monitoring mechanisms ensures long-term resilience against evolving threats.

Conclusion

The **Cybersecurity Internship Final Project** successfully demonstrated the full lifecycle of web application security hardening. Over six weeks, the project evolved from a vulnerable prototype to a secure, production-ready system. The experience provided hands-on exposure to real-world cybersecurity practices, including vulnerability assessment, ethical hacking, and system hardening.

The project outcomes align with **OWASP Top 10**, **NIST**, and **CIS Benchmarks**, showcasing a strong understanding of both offensive and defensive security principles.

Recommendations

1. Implement **HTTPS** with valid SSL/TLS certificates in production.
2. Conduct **monthly security audits** and vulnerability scans.
3. Regularly update **Apache, PHP, and OpenSSL** to the latest stable versions.
4. Deploy a **Web Application Firewall (WAF)** for additional protection.
5. Schedule **quarterly penetration testing** to ensure continuous security validation.

Final Remarks

This internship provided valuable experience in applying cybersecurity principles to real-world systems. The project not only enhanced technical proficiency but also reinforced the importance of proactive defense, continuous monitoring, and adherence to security best practices.

GitHub Repository:
https://github.com/mtalha27709-coder/user-management-security-hardening

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End of Report